

Fock Space Project

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We want to find the transform on the Fock space that is unitarily equivalent to the Hilbert transform on L^2 . We know that we can write Hf as

$$Hf = \mathcal{F}^{-1}(-i \operatorname{sgn}(\xi) \mathcal{F}(f)(\xi)),$$

where $\mathcal{F}$ is the Fourier transform (as an operator on $L^2(\mathbb{R})$) defined as:

$$\begin{aligned} (\mathcal{F}f)(x) &= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} f(t) e^{2ixt} dt \\ (\mathcal{F}^{-1}f)(x) &= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} f(t) e^{-2ixt} dt \end{aligned}$$

Define the operator U as $(Uf)(x) = -i \operatorname{sgn}(x) f(x)$. If we can find its unitary equivalent on the Fock space, we will be done as we know the operator unitarily equivalent to the Fourier transform:

Theorem 1. *If $\mathcal{F}$ and $\mathcal{F}^{-1}$ are respectively defined as above, then*

$$(B\mathcal{F}B^{-1})(f)(z) = f(iz) \quad \text{and} \quad (B\mathcal{F}^{-1}B^{-1})(f)(z) = f(-iz)$$

Proof. Let the n th Hermite function h_n be defined as follows, where H_n is the n th (physicist's) Hermite polynomial and $c = (2/\pi)^{1/4}$:

$$h_n(x) = \frac{c}{\sqrt{2^n n!}} e^{-x^2} H_n(\sqrt{2}x).$$

The statement of the theorem then follows from the following facts:

- h_n is an eigenfunction of $\mathcal{F}$ with eigenvalue i^n .
- The Hermite functions form an orthonormal basis for $L^2(\mathbb{R})$.
- The Bargmann transform of the n th Hermite function is $z^n / \sqrt{n!}$.

Thus, expand $f \in F^2$ as a Taylor series:

$$f(z) = \sum_{n=0}^{\infty} a_n z^n$$

Then, (where $\tilde{a}_n = a_n \sqrt{n!}$)

$$B\mathcal{F}B^{-1}(f)(z) = B\mathcal{F} \left(\sum_{n=0}^{\infty} \tilde{a}_n h_n(x) \right)$$

$$\begin{aligned}
&= B \left(\sum_{n=0}^{\infty} \tilde{a}_n(i)^n h_n(x) \right) \\
&= \sum_{n=0}^{\infty} a_n i^n z^n \\
&= f(iz)
\end{aligned}$$

Similarly we can get that $(B\mathcal{F}^{-1}B^{-1})(f)(z) = f(-iz)$. □

Now we find the operator $BUB^{-1} : F^2 \rightarrow F^2$:

Lemma 1.

$$(BUB^{-1})(f)(z) = -\frac{2i}{\sqrt{\pi}} \int_{\mathbb{C}} f(w) e^{z\bar{w}} A_1 \left(\frac{z + \bar{w}}{\sqrt{2}} \right) d\lambda(w),$$

where $A_1(z) = \int_0^z e^{-u^2} du$, i.e. the antiderivative of e^{-z^2} which is 0 at $z = 0$.

Proof.

$$\begin{aligned}
(BUB^{-1})(f)(z) &= -ci \int_{\mathbb{R}} e^{2xz - x^2 - z^2/2} \operatorname{sgn} x \int_{\mathbb{C}} f(w) e^{2x\bar{w} - x^2 - \bar{w}^2} d\lambda(w) dx \\
&= -ci \int_{\mathbb{C}} e^{-z^2/2 - \bar{w}^2} f(w) \int_{\mathbb{R}} \operatorname{sgn}(x) e^{-2x^2 + 2x(z + \bar{w})} dx d\lambda(w) \\
&= -ci \int_{\mathbb{C}} e^{z\bar{w}} f(w) \int_{\mathbb{R}} \operatorname{sgn}(x) e^{-(\sqrt{2}x - (z + \bar{w})/\sqrt{2})^2} dx d\lambda(w).
\end{aligned}$$

(Here $c = \sqrt{2/\pi}$).

Consider the entire function $h(z) = \int_{\mathbb{R}} \operatorname{sgn}(x) e^{-(\sqrt{2}x + z)^2} dx$. Note that $h(0) = 0$. Differentiating under the integral (which can be easily justified using DCT by noting that $e^{-\operatorname{Re}((\sqrt{2}x + z)^2)}$ is integrable), we get

$$\begin{aligned}
h'(z) &= -2 \int_{\mathbb{R}} \operatorname{sgn} x (\sqrt{2}x + z) e^{-(\sqrt{2}x + z)^2} dx \\
&= 2 \int_{-\infty}^0 (\sqrt{2}x + z) e^{-(\sqrt{2}x + z)^2} dx - 2 \int_0^{\infty} (\sqrt{2}x + z) e^{-(\sqrt{2}x + z)^2} dx \\
&= \frac{1}{\sqrt{2}} \left[- \left[e^{-(\sqrt{2}x + z)^2} \right]_{-\infty}^0 + \left[e^{-(\sqrt{2}x + z)^2} \right]_0^{\infty} \right] \\
&= -\sqrt{2} e^{-z^2}.
\end{aligned}$$

Hence we have that $h(z) = -\sqrt{2}A_1(z)$. Thus,

$$\begin{aligned}
(BUB^{-1})(f)(z) &= \sqrt{2}ci \int_{\mathbb{C}} f(w) e^{z\bar{w}} A_1 \left(-\frac{z + \bar{w}}{\sqrt{2}} \right) d\lambda(w) \\
&= -\frac{2i}{\sqrt{\pi}} \int_{\mathbb{C}} f(w) e^{z\bar{w}} A_1 \left(\frac{z + \bar{w}}{\sqrt{2}} \right) d\lambda(w).
\end{aligned}$$

□

Lemma 2. Let $A(z) = \int_0^z e^{u^2} du$, i.e. the antiderivative of e^{z^2} which is 0 at $z = 0$. Then,

$$A_1(iz) = iA(z) \quad \text{and} \quad A(iz) = iA_1(z)$$

Proof. This follows directly by a change of variables. $\square$

Theorem 2.

$$(Tf)(z) := (BHB^{-1})(f)(z) = -\frac{2}{\sqrt{\pi}} \int_{\mathbb{C}} f(w) e^{z\bar{w}} A\left(\frac{z-\bar{w}}{\sqrt{2}}\right) d\lambda(w)$$

Proof. Now we find the transformation unitarily equivalent to the Hilbert transform. Since $(B\mathcal{F}B^{-1})(f)(z) = f(iz)$, we have that

$$(BUB^{-1})(B\mathcal{F}B^{-1})(f)(z) = -\frac{2i}{\sqrt{\pi}} \int_{\mathbb{C}} f(iw) e^{z\bar{w}} A_1\left(\frac{z+\bar{w}}{\sqrt{2}}\right) d\lambda(w).$$

Now since $(B\mathcal{F}^{-1}B^{-1})(f)(z) = f(-iz)$, we get

$$(BHB^{-1})(f)(z) = -\frac{2i}{\sqrt{\pi}} \int_{\mathbb{C}} f(iw) e^{-iz\bar{w}} A_1\left(\frac{-iz+\bar{w}}{\sqrt{2}}\right) d\lambda(w).$$

Doing a change of variable of $iw \mapsto w$,

$$(BHB^{-1})(f)(z) = -\frac{2i}{\sqrt{\pi}} \int_{\mathbb{C}} f(w) e^{z\bar{w}} A_1\left(\frac{-iz+i\bar{w}}{\sqrt{2}}\right) d\lambda(w).$$

Now using the second lemma and that A is an odd function, we get that

$$(BHB^{-1})(f)(z) = -\frac{2}{\sqrt{\pi}} \int_{\mathbb{C}} f(w) e^{z\bar{w}} A\left(\frac{z-\bar{w}}{\sqrt{2}}\right) d\lambda(w).$$

$\square$

Next we identify a family of functions in F^2 which diagonalize the operator T , using its unitary equivalence to the Hilbert transform.

Theorem 3. The family of functions $\phi_n : F^2 \rightarrow F^2$ defined as

$$\phi_n(z) = \frac{ie^{z^2/2}}{n!} \frac{\partial^n}{\partial t^n} \left[\frac{e^{-(z+\frac{i}{2}(\frac{1+t}{1-t}))^2} A\left(z+\frac{i}{2}(\frac{1+t}{1-t})\right)}{1-t} \right]_{t=0}$$

diagonalize the operator T .

Proof. It is known that there is an orthonormal basis ψ_n for $L^2(\mathbb{R})$ which diagonalizes the Hilbert transform, namely

$$\psi_n(x) = \sqrt{\frac{2}{\pi}} (2ix-1)^{-1} \left(\frac{2ix+1}{2ix-1} \right)^n$$

These functions also have known (inverse) Fourier transforms:

$$\hat{\psi}_n(x) = L_n(2x) e^{-x} \chi_{[0,\infty)}(x)$$

for $n = 0, 1, 2, \dots$ where L_n is the n th Laguerre polynomial of type 0. L_n can be defined by the generating function:

$$\sum_{n=1}^{\infty} L_n(x) t^n = \frac{e^{-\frac{xt}{1-t}}}{1-t}$$

Therefore,

$$\sum_{n=1}^{\infty} \hat{\psi}_n(x) t^n = \sum_{n=1}^{\infty} L_n(2x) e^{-x} \chi_{[0,\infty)} t^n = \frac{e^{-\frac{2xt}{1-t} - x}}{1-t} \chi_{[0,\infty)} = \frac{e^{-x \frac{1+t}{1-t}}}{1-t} \chi_{[0,\infty)}.$$

Taking the Bargmann transform on both sides,

$$\begin{aligned} \sum_{n=1}^{\infty} (B\hat{\psi}_n)(z) t^n &= \frac{1}{1-t} B \left(e^{-x \frac{1+t}{1-t}} \chi_{[0,\infty)} \right) (z) \\ &= \frac{1}{1-t} e^{-\frac{z^2}{2}} e^{(z - \frac{1}{2}(\frac{1+t}{1-t}))^2} A_1 \left(z - \frac{1}{2} \left(\frac{1+t}{1-t} \right) \right), \end{aligned}$$

using the next lemma. The proof is complete by noting that $(B\psi_n)(z) = (B\hat{\psi}_n)(iz)$ from before, as $\hat{\psi} = \mathcal{F}^{-1}\psi$. □

Lemma 3. For fixed $a \in \mathbb{C}$, if $f(x) = e^{-ax} \chi_{[0,\infty)}$,

$$(Bf)(z) = e^{\frac{z^2}{2} + \frac{a^2}{4} - az} A_1 \left(z - \frac{a}{2} \right) = e^{-\frac{z^2}{2}} e^{(z - a/2)^2} A_1 \left(z - \frac{a}{2} \right).$$

Proof.

$$\begin{aligned} (Bf)(z) &= \int_0^{\infty} e^{-ax} e^{-x^2 + 2xz - \frac{z^2}{2}} dx \\ &= e^{\frac{z^2}{2} + \frac{a^2}{4} - az} \int_0^{\infty} e^{-(x - (z - \frac{a}{2}))^2} dx \\ &= e^{\frac{z^2}{2} + \frac{a^2}{4} - az} A_1 \left(z - \frac{a}{2} \right), \end{aligned}$$

by a similar calculation as in Lemma 1. □

We now calculate explicitly the action of T on the orthonormal basis $\{z^n / \sqrt{n!}\}$.

Lemma 4. Let $\tilde{e}_n(w) = w^n$. Then the following recurrence relation holds:

$$(T\tilde{e}_{n+1})(z) = z(T\tilde{e}_n)(z) + n(T\tilde{e}_{n-1})(z) - (T\tilde{e}_n)'(z). \quad (1)$$

In particular, T satisfies the following commutativity relation when f is a monomial:

$$T(f' - wf)(z) = (Tf)'(z) - z(Tf)(z)$$

Proof. Note that using the power series expansion, we have

$$g(w) := A \left(\frac{z-w}{\sqrt{2}} \right) e^{zw} = \sum_{n=0}^{\infty} g^{(n)}(0) \frac{w^n}{n!},$$

so that $(T\tilde{e}_n)(z) = g^{(n)}(0)$, using orthogonality of monomials. We first evaluate the third term on the RHS of (1) using the Cauchy integral formula, where γ is the unit circle around the origin:

$$\frac{d}{dz} \left(g^{(n)}(0) \right) = \frac{n!}{2\pi i} \int_{\gamma} \frac{d}{dz} A \left(\frac{z-w}{\sqrt{2}} \right) e^{zw} \frac{1}{w^{n+1}} dw$$

$$\begin{aligned}
&= \frac{n!}{2\pi i} \int_{\gamma} \frac{wA\left(\frac{z-w}{\sqrt{2}}\right) e^{zw} + \frac{1}{\sqrt{2}} e^{\frac{1}{2}(z-w)^2} e^{zw}}{w^{n+1}} dw \\
&= ng^{(n-1)}(0) + \frac{n!}{2\sqrt{2}\pi i} \int_{\gamma} \frac{e^{\frac{1}{2}(z-w)^2} e^{zw}}{w^{n+1}} dw.
\end{aligned}$$

Now consider the complete RHS of (1):

$$\begin{aligned}
zg^{(n)}(0) + ng^{(n-1)}(0) - \frac{d}{dz}g^{(n)}(0) &= \frac{n!}{2\pi i} \int_{\gamma} \frac{1}{w^{n+1}} \left[zA\left(\frac{z-w}{\sqrt{2}}\right) e^{zw} - \frac{e^{\frac{1}{2}(z-w)^2} e^{zw}}{\sqrt{2}} \right] dw \\
&= \frac{n!}{2\pi i} \int_{\gamma} \frac{g'(w)}{w^{n+1}} dw \\
&= g^{(n+1)}(0),
\end{aligned}$$

as claimed. □

Theorem 4.

$$T\tilde{e}_n(z) = \sum_{j=0}^n \binom{n}{j} (-1)^j z^{n-j} (T\tilde{e}_0)^{(j)}(z),$$

$$\text{and } (T\tilde{e}_0)(z) = -\frac{2}{\sqrt{\pi}} A\left(\frac{z}{\sqrt{2}}\right).$$

Proof. First we calculate $(T\tilde{e}_0)(z)$:

$$(T\tilde{e}_0)(z) = -\frac{2}{\sqrt{\pi}} \int_{\mathbb{C}} e^{z\bar{w}} A\left(\frac{z-\bar{w}}{\sqrt{2}}\right) d\lambda(w).$$

Make the substitution $w \rightarrow \bar{w}$ to get

$$(T\tilde{e}_0)(z) = -\frac{2}{\sqrt{\pi}} \int_{\mathbb{C}} e^{zw} A\left(\frac{z-w}{\sqrt{2}}\right) d\lambda(w).$$

Using polar coordinates and that the integrand is now an entire function of w , we get that

$$\begin{aligned}
(T\tilde{e}_0)(z) &= -\frac{2}{\sqrt{\pi}} e^{zw} A\left(\frac{z-w}{\sqrt{2}}\right) \Big|_{w=0} \\
&= -\frac{2}{\sqrt{\pi}} A\left(\frac{z}{\sqrt{2}}\right).
\end{aligned}$$

We use induction and the recurrence of the previous lemma to prove the claim, which is trivially true for $n = 0$. The $n = 1$ case follows by using (1). Assume $n > 0$ and that the claim is true for all $k \leq n$. Then,

$$\begin{aligned}
z(T\tilde{e}_n)(z) &= \sum_{j=0}^n \binom{n}{j} (-1)^j z^{n-j+1} (T\tilde{e}_0)^{(j)}(z) \\
n(T\tilde{e}_{n-1})(z) &= n \cdot \sum_{j=0}^{n-1} \binom{n-1}{j} (-1)^j z^{n-j-1} (T\tilde{e}_0)^{(j)}(z) \\
-(T\tilde{e}_n)'(z) &= \sum_{j=0}^n \binom{n}{j} (-1)^{j+1} [(n-j)z^{n-j-1} (T\tilde{e}_0)^{(j)}(z) + z^{n-j} (T\tilde{e}_0)^{(j+1)}]
\end{aligned}$$

$$= n \cdot \sum_{j=0}^{n-1} \binom{n-1}{j} (-1)^{j+1} z^{n-j-1} (T\tilde{e}_0)^{(j)}(z) + \sum_{j=0}^n \binom{n}{j} (-1)^{j+1} z^{n-j} (T\tilde{e}_0)^{(j+1)}(z).$$

The first term of $-(T\tilde{e}_n)'(z)$ cancels with $n(T\tilde{e}_{n-1})(z)$. The second term can be written as

$$\sum_{j=0}^n \binom{n}{j} (-1)^{j+1} z^{n-j} (T\tilde{e}_0)^{(j+1)}(z) = \sum_{j=1}^{n+1} \binom{n}{j-1} (-1)^j z^{n-j+1} (T\tilde{e}_0)^{(j)}(z).$$

Now adding the three equations above,

$$\begin{aligned} \sum_{j=1}^n (-1)^j z^{n-j+1} (T\tilde{e}_0)^{(j)}(z) \left[\binom{n}{j} + \binom{n}{j-1} \right] + z^{n+1} (T\tilde{e}_0)(z) + (-1)^{n+1} (T\tilde{e}_0)^{(n+1)}(z) \\ = \sum_{j=0}^{n+1} \binom{n+1}{j} (-1)^j z^{n+1-j} (T\tilde{e}_0)^{(j)}(z), \end{aligned}$$

hence proving the claim. We have used simple combinatorial identities throughout. $\square$

Note that $(-1)^j (T\tilde{e}_0)^{(j)}(z)$ is closely related to the Hermite functions (i.e. as functions on F^2):

$$\begin{aligned} (-1)^j (T\tilde{e}_0)^{(j)}(z) &= (-1)^j \frac{d^j}{dz^j} \left(-\frac{2}{\sqrt{\pi}} A \left(\frac{z}{\sqrt{2}} \right) \right) \\ &= (-1)^{j-1} \frac{2}{\sqrt{\pi}} \frac{d^{j-1}}{dz^{j-1}} \left(\frac{1}{\sqrt{2}} e^{z^2/2} \right) \\ &= \sqrt{\frac{2}{\pi}} i^{j-1} e^{z^2/2} H_{j-1}(iz) \end{aligned}$$

where H_n is the n th (probabilist's) Hermite polynomial.

Once we prove that T is a bounded (in fact unitary) operator in the next section, the commutativity relation of T mentioned in Lemma 4 for monomials can be extended for all f .

Unitarity of T

We will prove the unitarity and boundedness of T directly by showing that T satisfies $T^2 = -\text{Id}$. It is clear already that $T^* = -T$ by noting that $\overline{h(z, w)} = -h(w, z)$, where $h(z, w) = e^{\bar{z}w} A \left(\frac{\bar{z}-w}{\sqrt{2}} \right)$, from which the unitarity of T follows.

The desired result will follow immediately from the following theorem, which appears interesting in itself.

Theorem 5.

$$\int_{\mathbb{C}} A \left(\frac{u - \bar{z}}{\sqrt{2}} \right) e^{\bar{z}u} A \left(\frac{\bar{u} - w}{\sqrt{2}} \right) e^{w\bar{u}} d\lambda(u) = \frac{\pi}{4} e^{w\bar{z}}$$

In particular, for fixed $w \in \mathbb{C}$, $A((z - \bar{w})/\sqrt{2}) e^{z\bar{w}} \in F^2$.

Proof. We will show that the above equality holds on the diagonal of $\mathbb{C}^2$, i.e. when $z = w$. Using a theorem from several complex variables, this will establish the equality on all of $\mathbb{C}^2$. So, we have to show that

$$I(z) := \int_{\mathbb{C}} A \left(\frac{u - \bar{z}}{\sqrt{2}} \right) e^{\bar{z}u} A \left(\frac{\bar{u} - z}{\sqrt{2}} \right) e^{z\bar{u}} d\lambda(u) = \frac{\pi}{4} e^{z\bar{z}}.$$

Do the change of variable $u \mapsto u + z$ to get

$$I(z) = e^{z\bar{z}} \int_{\mathbb{C}} A\left(\frac{u+z-\bar{z}}{\sqrt{2}}\right) A\left(\frac{\bar{u}+\bar{z}-z}{\sqrt{2}}\right) d\lambda(u)$$

Now $z - \bar{z}$ is a purely imaginary number, say it . The proof is then completed by the following lemma. $\square$

Lemma 5. *The following holds for all $t \in \mathbb{R}$:*

$$\int_{\mathbb{C}} A\left(\frac{u+it}{\sqrt{2}}\right) A\left(\frac{\bar{u}-it}{\sqrt{2}}\right) d\lambda(u) = \frac{\pi}{4}$$

Proof. Let $f(t)$ denote the left hand side of the claim. We will prove that f is a constant function by showing $f'(t) = 0$:

$$\begin{aligned} f'(t) &= \frac{d}{dt} \int_{\mathbb{C}} A\left(\frac{u+it}{\sqrt{2}}\right) A\left(\frac{\bar{u}-it}{\sqrt{2}}\right) d\lambda(u) \\ &= \frac{i}{\sqrt{2}} \int_{\mathbb{C}} \left[e^{\frac{1}{2}(u+it)^2} A\left(\frac{\bar{u}-it}{\sqrt{2}}\right) - e^{\frac{1}{2}(\bar{u}-it)^2} A\left(\frac{u+it}{\sqrt{2}}\right) \right] d\lambda(u) \end{aligned}$$

Now use the change of variable $u \mapsto -\bar{u}$ on the second term inside square brackets, and that A is an odd function, to conclude that the right hand side is 0. Hence $f(t)$ is a constant. We will now calculate its value at $t = 0$ to finish the proof (modulo justification of differentiation under the integral sign):

$$f(0) = \int_{\mathbb{C}} A\left(\frac{u}{\sqrt{2}}\right) A\left(\frac{\bar{u}}{\sqrt{2}}\right) d\lambda(u)$$

This is immediately recognized as the squared norm in F^2 of the function $A\left(\frac{u}{\sqrt{2}}\right)$, which can be calculated from its Taylor expansion:

$$\begin{aligned} A\left(\frac{u}{\sqrt{2}}\right) &= \sum_{n=0}^{\infty} \frac{(u/\sqrt{2})^{2n+1}}{(2n+1)n!} \\ &= \sum_{n=0}^{\infty} \frac{u^{2n+1}}{2^n \sqrt{2} (2n+1)n!} \\ &= \sum_{n=0}^{\infty} \underbrace{\frac{\sqrt{(2n+1)!}}{2^n \sqrt{2} (2n+1)n!}}_{a_n} \frac{u^{2n+1}}{\sqrt{(2n+1)!}} \end{aligned}$$

Since $u^n / \sqrt{n!}$ is an orthonormal basis, we have

$$\left\| A\left(\frac{u}{\sqrt{2}}\right) \right\|^2 = \sum_{n=0}^{\infty} a_n^2 = \frac{1}{2} \sum_{n=0}^{\infty} \frac{(2n+1)!}{4^n (2n+1)^2 (n!)^2} = \frac{1}{2} \sum_{n=0}^{\infty} \binom{2n}{n} \frac{1}{4^n (2n+1)}$$

This last sum is the series expansion of the arcsine function evaluated at 1. Thus, we have

$$f(0) = \left\| A\left(\frac{u}{\sqrt{2}}\right) \right\|^2 = \frac{1}{2} \arcsin(1) = \frac{\pi}{4}.$$

$\square$

Theorem 6.

$$T(Tf)(z) = -f(z)$$

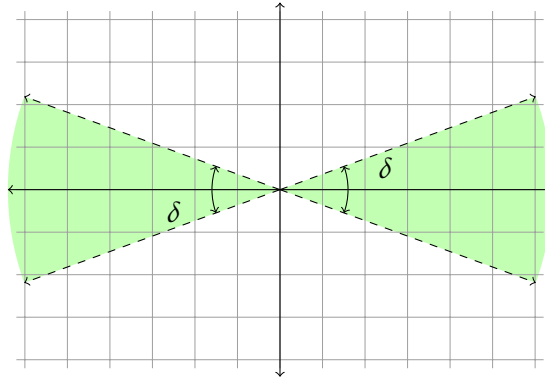
The proof only requires a justification for applying Fubini's theorem. Unfortunately we cannot apply Fubini's theorem directly; we show that the integrals can be interchanged when integrating over a smaller set and then use a limiting argument. For this we need a lemma:

Lemma 6. *Let $\delta > 0$, and $S_\delta = \{z \mid \arg(z) \notin (0, \delta) \cup (\pi - \delta, \pi + \delta) \cup (2\pi - \delta, 2\pi)\}$. Then,*

$$\int_{\mathbb{C}} |z^n| \int_{S_\delta} \left| A\left(\frac{u - \bar{z}}{\sqrt{2}}\right) e^{\bar{z}u} A\left(\frac{\bar{u} - w}{\sqrt{2}}\right) e^{w\bar{u}} \right| d\lambda(u) d\lambda(z) < \infty,$$

i.e. over the smaller domain of integration S_δ , the order of integration can be switched.

Figure 1: S_δ is the area *excluding* the green shaded region



Proof. From the mean value theorem for vector valued functions, we get

$$\begin{aligned} \left| A\left(\frac{z}{\sqrt{2}}\right) \right| &\leq |z| \cdot \sup_{t \in [0,1]} |e^{t^2 z^2 / 2}| = \begin{cases} |z| \cdot |e^{z^2 / 2}| & \operatorname{Re}(z^2) > 0 \\ |z| & \operatorname{Re}(z^2) \leq 0 \end{cases} \\ &\leq |z| \cdot (|e^{z^2 / 2}| + 1) \end{aligned}$$

We use this inequality on the inner integral:

$$\begin{aligned} \int_{S_\delta} \left| A\left(\frac{u - \bar{z}}{\sqrt{2}}\right) e^{\bar{z}u} A\left(\frac{\bar{u} - w}{\sqrt{2}}\right) e^{w\bar{u}} \right| d\lambda(u) &\leq \\ \int_{S_\delta} \left(\left| e^{\frac{1}{2}(u - \bar{z})^2} \right| + 1 \right) |e^{\bar{z}u}| \left(\left| e^{\frac{1}{2}(\bar{u} - w)^2} \right| + 1 \right) |e^{w\bar{u}}| |u - \bar{z}| |\bar{u} - w| d\lambda(u) \end{aligned}$$

On expanding the two parenthesized expressions, we get the following expression:

$$\left| e^{\frac{1}{2}(u - \bar{z})^2 + \frac{1}{2}(\bar{u} - w)^2} \right| + \left| e^{\frac{1}{2}(u - \bar{z})^2} \right| + \left| e^{\frac{1}{2}(\bar{u} - w)^2} \right| + 1.$$

It is easy to see that the integrals corresponding to the latter three summands are finite, so we will focus on showing that the integral with the first summand is finite.

$$\int_{S_\delta} \left| e^{\frac{1}{2}(u - \bar{z})^2 + \frac{1}{2}(\bar{u} - w)^2 + \bar{z}u + w\bar{u}} \right| |u - \bar{z}| |\bar{u} - w| d\lambda(u) = \int_{S_\delta} \left| e^{\frac{1}{2}(u^2 + \bar{u}^2 + z^2 + w^2)} \right| |u - \bar{z}| |\bar{u} - w| d\lambda(u)$$

$$= \left| e^{\frac{1}{2}(z^2+w^2)} \right| \cdot \int_{\mathbb{C}} \left| e^{\frac{1}{2}(u^2+\bar{u}^2)} \right| |u - \bar{z}| |\bar{u} - w| d\lambda(u) \quad (2)$$

Now we use the triangle inequality and switch to polar coordinates (ignoring the factor outside the integral for now):

$$\begin{aligned} \int_{S_\delta} \left| e^{\frac{1}{2}(u^2+\bar{u}^2)} \right| |u - \bar{z}| |\bar{u} - w| d\lambda(u) &\leq \int_{S_\delta} \left| e^{\operatorname{Re}(u^2)} \right| (|u| + |\bar{z}|)(|\bar{u}| + |w|) d\lambda(u) \\ &= \int_{S_\delta} e^{\operatorname{Re}(u^2)} (|u| + |\bar{z}|)(|\bar{u}| + |w|) e^{-|u|^2} dA(u) \\ &= 2 \int_0^\infty (r + |\bar{z}|)(r + |w|) r \int_\delta^{\pi-\delta} e^{-2r^2 \sin^2(\theta)} d\theta dr \end{aligned}$$

This integral is finite and is a polynomial in $|z|$ and $|w|$, since $\sin(\theta)$ is bounded away from 0 on $(\delta, \pi - \delta)$.

From here it is clear that on evaluating the outer integral (including the ignored factor in (2)) we will again be left with a finite quantity, proving the lemma. $\square$

Proof of Theorem. The lemma implies that

$$\begin{aligned} \int_{\mathbb{C}} \int_{S_\delta} z^n A\left(\frac{u - \bar{z}}{\sqrt{2}}\right) e^{\bar{z}u} A\left(\frac{\bar{u} - w}{\sqrt{2}}\right) e^{w\bar{u}} d\lambda(u) d\lambda(z) = \\ \int_{S_\delta} \int_{\mathbb{C}} z^n A\left(\frac{u - \bar{z}}{\sqrt{2}}\right) e^{\bar{z}u} A\left(\frac{\bar{u} - w}{\sqrt{2}}\right) e^{w\bar{u}} d\lambda(z) d\lambda(u), \quad (3) \end{aligned}$$

i.e. the order of the two integrals can be swapped, for every $\delta > 0$. We want to take the limit as $\delta \downarrow 0$.

We use that the iterated integrals (when integrating over $\mathbb{C}$ instead of S_δ on both sides) exist. For the right side this follows from T being an operator into F^2 . For the left side, the inner integral exists because it is the inner product of two L^2 functions (see Theorem 5), and the outer integral's existence again follows from Theorem 5.

We use the following theorem from Royden's *Real Analysis, 4th Edition*:

Theorem. Let $(X, \mathcal{M}, \mu)$ be a measure space and f be integrable over X . If $\{X_n\}_{n=1}^\infty$ is an ascending countable collection of measurable subsets of X whose union is X , then

$$\int_X f d\mu = \lim_{n \rightarrow \infty} \int_{X_n} f d\mu.$$

This theorem, along with the fact that $\cup_{\delta \in \mathbb{Q}^+} S_\delta = \mathbb{C}$ and that both iterated integrals exists implies that

$$\begin{aligned} \lim_{\delta \downarrow 0} \int_{S_\delta} \int_{\mathbb{C}} z^n A\left(\frac{u - \bar{z}}{\sqrt{2}}\right) e^{\bar{z}u} A\left(\frac{\bar{u} - w}{\sqrt{2}}\right) e^{w\bar{u}} d\lambda(z) d\lambda(u) = \\ \int_{\mathbb{C}} \int_{\mathbb{C}} z^n A\left(\frac{u - \bar{z}}{\sqrt{2}}\right) e^{\bar{z}u} A\left(\frac{\bar{u} - w}{\sqrt{2}}\right) e^{w\bar{u}} d\lambda(z) d\lambda(u). \end{aligned}$$

From the fact that

$$\lim_{\delta \downarrow 0} \int_{S_\delta} A \left(\frac{u - \bar{z}}{\sqrt{2}} \right) e^{\bar{z}u} A \left(\frac{\bar{u} - w}{\sqrt{2}} \right) e^{w\bar{u}} d\lambda(u) = \int_{\mathbb{C}} A \left(\frac{u - \bar{z}}{\sqrt{2}} \right) e^{\bar{z}u} A \left(\frac{\bar{u} - w}{\sqrt{2}} \right) e^{w\bar{u}} d\lambda(u),$$

we know that there exists a δ_0 such that $\delta < \delta_0$ implies

$$\left| \int_{S_\delta} A \left(\frac{u - \bar{z}}{\sqrt{2}} \right) e^{\bar{z}u} A \left(\frac{\bar{u} - w}{\sqrt{2}} \right) e^{w\bar{u}} d\lambda(u) \right| \leq \left| \int_{\mathbb{C}} A \left(\frac{u - \bar{z}}{\sqrt{2}} \right) e^{\bar{z}u} A \left(\frac{\bar{u} - w}{\sqrt{2}} \right) e^{w\bar{u}} d\lambda(u) \right| + 1.$$

Since the right hand side is integrable, it serves as a dominating function for $\delta < \delta_0$ and so DCT is applicable.

We thus get that the order of integration can be switched even when the integration is over $\mathbb{C}$ and not S_δ , as required. $\square$